

Name: _____ Partner: _____ Period: _____

Tier R — Remediate

Directions: For each equation below, identify the type of conic, the center or vertex, and the key measurements. No equation-writing is required.

Card A: $y - 4 = 3(x + 2)^2$

Type: **parabola** Vertex: **$(-2, 4)$** $a = 3$ Opens: **up**

Card B: $\frac{(x - 3)^2}{25} + \frac{(y + 1)^2}{9} = 1$

Type: **ellipse** Center: **$(3, -1)$** $a = 5$ $b = 3$

Card C: $(x + 4)^2 + (y - 2)^2 = 36$

Type: **circle** Center: **$(-4, 2)$** $r = 6$

Card D: $\frac{(x - 1)^2}{9} - \frac{(y - 2)^2}{4} = 1$

Type: **hyperbola** Center: **$(1, 2)$** $a = 3$ $b = 2$ Opens: **left and right**

Teacher Note

Card D: the positive x -term means the hyperbola opens left and right. Students who say “up and down” are reading the wrong term as positive — redirect.

Tier A — Approaching Proficiency

Directions: Write the standard form equation for each card below.

Card A: Opens right, vertex $(-1, 2)$, passes through $(3, 4)$.

Use $x - h = a(y - k)^2$: $x - (-1) = a(y - 2)^2$. Substitute $(3, 4)$: $3 + 1 = a(4 - 2)^2 \Rightarrow 4 = 4a \Rightarrow a = 1$.

$x + 1 = (y - 2)^2$

Card B: Ellipse, center $(0, 0)$, $a = 4$ (horizontal), $b = 3$ (vertical).

$\frac{x^2}{16} + \frac{y^2}{9} = 1$

Card C: Hyperbola, center $(1, 0)$, $a = 3$, $b = 2$, opens left/right.

$\frac{(x - 1)^2}{9} - \frac{y^2}{4} = 1$

Card D: Circle, center $(-2, 5)$, radius 3.

$(x + 2)^2 + (y - 5)^2 = 9$

Teacher Note

Card A: students may try $y - k = a(x - h)^2$ (the wrong orientation). Redirect: “The problem says ‘opens to the right,’ so which variable is squared?” Card C: center is $(1, 0)$, so $k = 0$ and the y -term has no shift.

Tier E — Extension

Part 1. x -intercepts and y -intercepts of $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

x -intercepts (set $y = 0$): $\frac{x^2}{16} = 1 \Rightarrow x = \pm 4$. **x -intercepts: $(-4, 0)$ and $(4, 0)$**

y -intercepts (set $x = 0$): $\frac{y^2}{9} = 1 \Rightarrow y = \pm 3$. **y -intercepts: $(0, -3)$ and $(0, 3)$**

Part 2. Asymptotes of $\frac{(x-1)^2}{9} - \frac{y^2}{4} = 1$.

Center $(1, 0)$, $a = 3$, $b = 2$. **$y = \frac{2}{3}(x-1)$ and $y = -\frac{2}{3}(x-1)$**

Part 3. Card E — student-created. Accept any valid conic with matching equation.

Answers will vary. Check: the stated features match the equation, the correct standard form is used, and the equation is fully written.

Teacher Note

Part 3 is a creativity and synthesis check. Require students to verify their equation reproduces the features they stated (substitute the center/vertex into the equation to confirm it works).